

Predicting Customer Churn in the Telecommunication Industries Using Machine Learning Techniques

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Introduction

- Customer churn is a critical challenge that threatens profitability, sustainability, and competitiveness.
- Globally, churn rates in telecommunications range between 20–40%, leading to substantial revenue losses.
- Retaining customers is five to seven times cheaper than acquiring new ones (Kotler et al., 2021).
- In Sub-Saharan Africa, telecommunications access drives digital inclusion and economic participation (GSMA, 2022).

Problem Statement

- The Kenyan telecommunication market faces intensifying competition and rising expectations, elevating churn risk (Khan 2024).
- Many operators still manage churn reactively with static indicators and manual analysis (Jahromi 2020).
- These approaches are time-consuming, inefficient, and prone to human bias (Chen 2014).
- They miss real-time behavioral shifts and broader socio-economic changes, causing revenue leakage (Khan 2024).

Research Objectives

- i. To identify key churn predictors in telecommunication industry.
- ii. To develop a machine learning based churn prediction model using Kenyan telecommunications data.
- iii. To evaluate the performance of multiple Machine Learning algorithms.
- iv. To deploy an interactive dashboard integrated into telecommunication providers' systems.

Literature Review : Theoretical Framework

- Customer churn is a major challenge in telecommunications, reducing revenue and raising acquisition costs (Chen 2014). It underscores the economic impact of churn but lacks dynamic modeling.
- Churn management within Customer Relationship Management seeks to extend customer lifetime value (Nagarkar 2022). It links churn to strategic initiatives, though it remains largely reactive.
- Technology adoption theories explain retention and defection patterns (Shaikh 2024). They provide useful behavioral insights but are seldom integrated into churn models.

Literature Review : Service Industries and Early Models

- Service industries show churn driven by pricing, service quality, and customer experience (Smith 2021). This captures multiple drivers but is context-specific.
- Ensemble models such as Random Forest and Logistic Model Trees outperform single classifiers (Usman 2022). They raise accuracy but reduce interpretability.
- Deep learning models like CNNs achieve strong accuracy–profitability trade-offs (Saha 2023). They extract complex features but are resource-intensive.

Literature Review : Market-Specific Insights

- Market-specific modelling with XGBoost captures recharge frequency and inactivity (Nagarkar2022). It tailors strategies but often lacks clear metrics.
- Social Network Analysis features improve churn prediction AUC from 0.84 to 0.933 . They add relational value but are data-heavy (Ahmad2019).
- Temporal modelling captures win-back potential from usage and payment history (Phua2012). It offers longitudinal insight but is complex to apply.

Literature Review: Advanced Methods

- Transfer learning converts telecom data into image formats for CNNs. It is innovative but difficult to interpret (Ahmed 2019).
- Adaptive ensembles combining XGBoost, LightGBM, and LSTM reach up to 99.28% accuracy (Shaikh 2024). They achieve state-of-the-art results but act as black boxes.
- Profit-sensitive evaluation aligns churn models with business Return on Investment (Khan2024). It adds practical relevance but is rarely applied in African contexts.

- **Underutilization of Unstructured Data:**

Most churn models ignore text logs, complaint notes, and social media sentiment, missing experience-related signals and early indicators of defection.

- **Geographic Underrepresentation (Africa):**

Few studies are calibrated to African telecom markets particularly rural, low-resource contexts which limiting external validity and actionable retention insights.

Methodology: CRISP-DM Approach

- This study adopts the CRISP-DM framework, ensuring a structured, repeatable, and scalable process for churn prediction.
- Data will be collected via structured questionnaires administered to staff across multiple departments of a Kenyan telecommunication operator.
- Questionnaires will capture demographics, contracts, billing, service usage, and departmental perspectives on churn drivers.
- All responses will be anonymized and aligned with confidentiality and governance guidelines.

Methodology: Data Understanding and Cleaning

- A preliminary Exploratory Data Analysis (EDA) will summarize churn vs. non-churn patterns using statistics, cross-tabs, and visualizations.
- Data cleaning will enhance quality by imputing missing values, correcting errors, and standardizing categorical/numeric variables.
- Outliers will be detected through statistical thresholds and visual diagnostics, then treated through capping, transformation, or removal.
- This preprocessing ensures a robust dataset for reliable predictive modeling.

- Churn prediction framed as a binary classification task: churn vs. non-churn.
- Comparative modelling with Logistic Regression, Decision Tree, Random Forest, LightGBM, Ensembles, and Logistic Regression + SMOTE.
- Feature engineering: create tenure categories, normalized spending ratios, and usage intensity scores.
- Feature scaling, selection, and encoding applied to improve model accuracy, efficiency, and interpretability.

Methodology: Evaluation and Deployment

- Models trained with an 80/20 stratified split, preserving churn class balance.
- Performance measured using Precision, Recall, F1-Score, and ROC-AUC, with Recall prioritized due to business cost of false negatives.
- Best model deployed enabling seamless integration with CRM systems.
- API deployment provides scalability and real-time churn probability outputs for targeted retention campaigns.

- **Technical Outcomes**

- High predictive performance across Recall, Precision, F1-Score, and ROC-AUC, ensuring accurate identification of at-risk customers.

- **Operational Outcomes**

- Proactive retention strategies that reduce revenue leakage by identifying high-risk customers before termination.
- Optimized marketing and resource allocation by focusing interventions on customers with the highest churn probability and business value.

- **Strategic Contribution**

- A deployable, real-world solution addressing the immediate need for churn reduction.